

The Connection Problem

An AI Healthcare Growth Analysis System Connecting
Evidence and Human Judgment

Most Executives Don't Have a Data Problem. They Have a Connection Problem.



Healthcare organizations generate valuable marketing, patient, and financial evidence every day. When that evidence lives in separate systems and departments, however, leadership can struggle to understand what's actually driving outcomes.

What Every Executive Secretly Wishes for is a Marketing Volume Knob.



THE WISH

Turn marketing up and produce predictable growth and revenue.

THE REALITY

MARKETING
Spend ↑↑



ADMISSIONS
Admissions
↓

Consider this pattern: admissions are declining even as marketing spend and inquiry volume rise. Marketing sees demand that isn't converting. Admissions sees inquiries it believes are lower quality. Each department has evidence. Neither has enough connected context to explain the outcome.

The First Question in AI Development Shouldn't be About AI.

It should be about the organization it serves.



Before deciding what an AI system should do, first understand how the organization actually operates: how people make decisions, how information moves between them, and where those connections break down.

WHAT WAS BUILT

The HEOS Evidence Engine

I built one real, functioning AI healthcare growth analysis and evaluation system designed to examine evidence across different parts of an organization, connect what those parts are seeing, and identify where human judgment should be focused.

HARBOR RIDGE

CONTROLLED OPERATING ENVIRONMENT



MARKETING · PATIENT · BILLING EVIDENCE

EVIDENCE ENGINE

EXAMINE

evidence

CONNECT

what parts see

FOCUS

human judgment

To test it, I built Harbor Ridge, a synthetic behavioral healthcare organization designed to give the Evidence Engine a coherent operating environment to analyze while keeping the underlying data and answers under my control.

Harbor Ridge is the organization.

The Evidence Engine is what examined it.

The Hardest Part of Testing AI Isn't Accepting Failure.

It's Resisting the Temptation to Manufacture Success.

01

ANSWERS

02

EVALUATION CRITERIA

03

SUCCESS RULES

FROZEN BEFORE RESULTS WERE KNOWN

SYSTEM CAN SEE

OPERATING EVIDENCE

SYSTEM CANNOT SEE

FROZEN ANSWERS

used later for evaluation

BLIND

Before the Evidence Engine analyzed either test scenario, the expected answers, the evaluation criteria, and the rules for success were already frozen. This wasn't only a test of the system. It was a constraint on how I could interpret whatever it produced.

WHEN THE RULES ACTUALLY MATTERED

While building one test scenario, a validation defect appeared. I prepared a command to search across different random seeds for another result, then stopped before running it. Instead, I corrected the underlying validation rule and left the frozen evaluation rules unchanged.

OBSERVED FINDING

≈\$251K

An estimated \$251,000 in additional collection opportunities if out-of-network claims had realized at the in-network rate.

REALIZATION RATE

IN-NETWORK



OUT-OF-NETWORK



IN-NETWORK

\$1,216,954.73 billed
\$812,530.95 collected

OUT-OF-NETWORK

\$688,137.27 billed
\$209,054.85 collected

Calculated counterfactual gap: \$250,397.60

MISS / MISS / CLEAN

Together with the observed finding, the formal results clarified both the Evidence Engine's current capabilities and where further refinement is needed.

SCENARIO 1

Planted deterioration in
paid-search inquiry quality

SCENARIO 2

Planted deterioration in
professional-outreach quality

HEALTHY BASELINE

Unaltered control environment

BENCHMARK RESULT

MISS

BENCHMARK RESULT

MISS

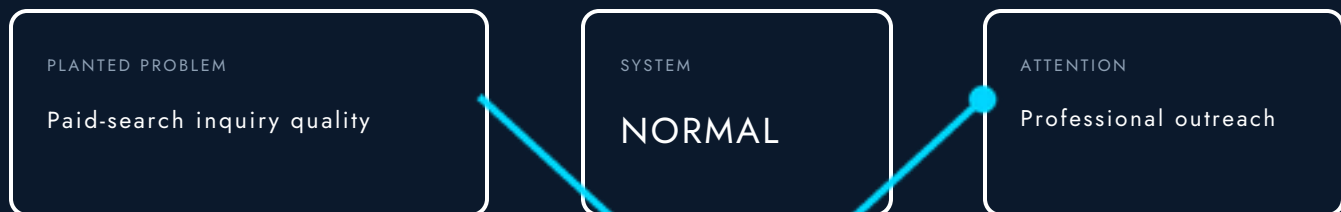
CONTROL RESULT

CLEAN

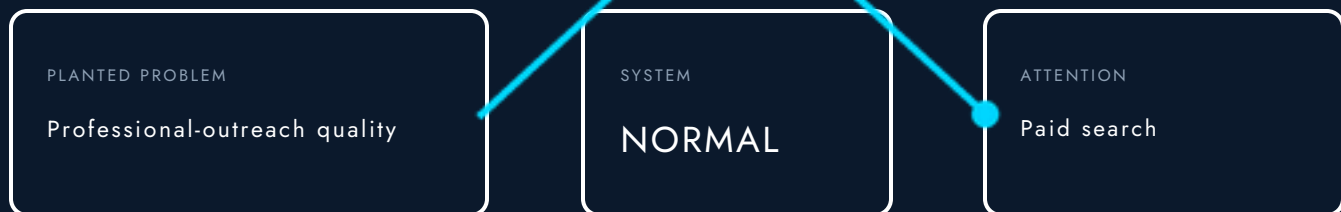
The Misses Were More Than Omissions

In both benchmark scenarios, the Evidence Engine did more than fail to identify the planted problem. It characterized the relevant domain as normal and directed analytical attention elsewhere.

SCENARIO 1



SCENARIO 2



For decision support, this distinction matters: failing to identify a problem is different from directing human attention away from the area where the problem actually exists.

The Findings Stay. So Do the Misses.

≈\$251K

OBSERVED FINDING

MISS / MISS

BENCHMARK RESULTS

CLEAN

CONTROL RESULT

The Evidence Engine surfaced a financially meaningful observed finding, missed both problems it was explicitly tested to find, and left the unaltered control clean. Each result answers a different question. None changes what the others establish.

- A useful finding is not the same as passing the benchmark.
- A benchmark miss does not mean nothing useful was found.
- A clean control is evidence of this evaluation, not proof of general reliability.

Where the Evidence Stops

The results are real. They also have boundaries. Four matter here.

01 CONTROLLED ENVIRONMENT

Harbor Ridge is a constructed environment, not a real healthcare organization. The results describe Evidence Engine performance within that controlled environment.

02 AUTONOMOUS DISCOVERY

Both planted problems were missed. Reliable autonomous discovery of hidden, localized problems has not yet been demonstrated.

03 GENERAL RELIABILITY

The Healthy Baseline remained CLEAN under the frozen control criteria. One clean control does not establish general reliability across other conditions.

04 CAUSE OF THE MISSES

The closing diagnostic points toward signal structure as one possible explanation for the misses, but discovery procedure also remains possible. The cause is not settled.

What Transfers Is the Process

Harbor Ridge is not the template. Its schema reflects one specific operating environment. What transfers is the discipline behind it: understand how an organization actually operates, build around that reality, test whether the resulting analysis deserves trust, then use it to focus executive judgment.

SAME DISCIPLINE · DIFFERENT REALITY

HARBOR RIDGE



ORTHOPEDIC PRACTICE



A different organization requires the same discipline applied to a different reality. An orthopedic practice, for example, might organize evidence around referral and authorization, evaluation and treatment, possible surgery, rehabilitation, and follow-up. Its evidence architecture should reflect those relationships, not Harbor Ridge's behavioral-health schema.

Building Toward Predictable Growth

The Evidence Engine demonstrated the ability to connect evidence across departments and surface a measurably valuable insight. It also showed what comes next: model a real practice, connect marketing activity to financial outcomes, test the system against known answers, and use what it gets right and wrong to refine the next iteration.

The goal is attribution leadership can trust: enough visibility to identify where growth is breaking down, focus human judgment on the right problem, intervene, and measure what changes.

01 Understand the Practice Before Choosing the Technology.

Map how it actually operates, how decisions are made, and where evidence lives before deciding what AI should do.

02 Connect Evidence Around the Outcome That Matters.

Start with the business outcome you're actually trying to understand, not everything a system could technically access.

03 Decide What Would Earn Trust Before Seeing the Answer.

Establish what would count as success before the results are known, rather than adjusting the standard afterward.

The goal isn't to remove executive judgment from the process. It's to give judgment better-connected evidence and a clearer place to focus.

About the Work

THE PROJECT

The HEOS Evidence Engine is an AI healthcare growth analysis and evaluation system built to examine evidence across different parts of an organization, connect what those parts are seeing, and identify where human judgment should be focused. It was built and evaluated using Harbor Ridge, a controlled synthetic operating environment.

KEN CORIOSO

My work sits at the intersection of healthcare operations, growth, analytics, and AI. I built the HEOS Evidence Engine to explore how those disciplines can work together to help healthcare leaders make better-informed decisions.

01 EXPLORE THE PROJECT

github.com/kencorioso/ai-healthcare-growth-systems

02 CONNECT ON LINKEDIN

linkedin.com/in/kenneth-corioso/